
SPSS Exam

Exercise 1 . (20 pts) A study was conducted to investigate how effective a new diet was in lowering cholesterol. Results for the randomly selected subjects are shown in the table. The differences have a normal distribution. Are the subjects' cholesterol levels lower on average after the diet ?

Subject	A	B	C	D	E	F	G	H	I
Before	209	210	205	198	216	217	238	240	222
After	199	207	189	209	217	202	211	223	201

Perform the appropriate test with $\alpha = .05$.

Find the appropriate confidence interval for the mean's difference.

What do you conclude about this study ?

Exercise 2 . (35 pts) A researcher measures how positive a person's mood is and how creative he or she is, obtaining the following interval scores :

<i>Creativity</i>	<i>Mood</i>
7	10
6	8
11	9
4	6
5	5
7	3
4	7
5	2
6	4
4	1

I) Find the mean, median, mode, quartiles, 56th percentile, variance, standard deviation, minimum, and maximum of 'creativity'.

Draw a histogram for 'mood'.

II) You want to know if 'creativity' can be predicted from 'mood'.

Prepare a scatter diagram in SPSS to study the relationship between 'creativity' and 'mood'

Check the assumptions for the correlation test, perform then interpret it.

Use linear regression in SPSS to find the 90 percent prediction interval for the slope.

Write the linear equation.

Exercise 3 . (45 pts) Here are data from an experiment studying the effect of age on creativity scores :

<i>Age 4</i>	<i>Age 6</i>	<i>Age 8</i>	<i>Age 10</i>
3	9	9	7
5	11	12	7
7	14	9	6
4	10	8	4
3	10	9	5

I) Perform the two independent samples test comparing only two samples of your choice out of the four samples.

Check the assumptions.

Find the confidence interval of the difference in means and interpret it.

What conclusions can you draw from these results?

II) Perform a whole hypothesis testing procedure for the four samples together to compare means.

What should you conclude about this relationship?

Interpret your results for the test as well the pairwise comparisons.

What conclusions can you draw from these results?